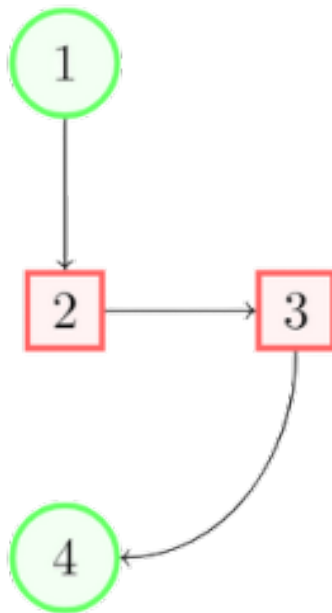


Drawing shapes using Tikz



November 5 2024

1 Including the tikz package for drawing shapes

To draw a shape in tikz, we need to use the `tikzpicture` environment.

Listing 1: The `tikzpicture` environment

```
\begin{tikzpicture}
    .....
\end{tikzpicture}
```

Now, in order to draw a line, we need to use the `\draw` command as follows:

Listing 2: Drawing a line using tikz

```
\draw (0,0)--(5,0);
```

This produces the following output:

The `;` after the command indicates end of the command. Now, we can include the `tikzpicture` within the `center` environment as follows:

Listing 3: Putting `tikzpicture` within `center` environment

```
\begin{center}
    \begin{tikzpicture}
        \draw (0,0)--(5,0);
    \end{tikzpicture}
\end{center}
```

This generates the following output:

The line is in the center. To change the thickness of the line, we can specify optional arguments with the `\draw` command. Below are some listings that demonstrate how to create lines of different thickness.

Listing 4: Thick line

```
\begin{center}
    \begin{tikzpicture}
        \draw [thick] (0,0)--(5,0);
    \end{tikzpicture}
\end{center}
```

```
\draw[thick] (0,0)--(5,0);
\end{tikzpicture}
\end{center}
```

The output produced by this listing is as follows:

We can draw lines of even more thicker than this one:

Listing 5: A thicker line

```
\begin{center}
\begin{tikzpicture}
\draw[very thick] (0,0)--(5,0);
\end{tikzpicture}
\end{center}
```

Produces this output:

If we want to specify our own value for thickness, then we can use the `line width` option with the `\draw` command like so:

Listing 6: Line of custom thickness

```
\begin{center}
\begin{tikzpicture}
\draw[line width=5mm] (0,0)--(5,0);
\end{tikzpicture}
\end{center}
```

This will produce the following output:



Notice that the ends of the line are sharp, we can smoothen them by using the `cap` option like so:

Listing 7: Rounding the end of the line

```
\begin{center}
\begin{tikzpicture}
\draw[line width=5mm, cap=round] (0,0)--(5,0);
\end{tikzpicture}
\end{center}
```

The following output is produced:



We can also change the color of the line and we don't have to use any option variables, we just need to provide the color name as an argument to `\draw` command like so:

Listing 8: Changing the color of the line

```
\begin{center}
  \begin{tikzpicture}
    \draw[line width=5mm, cap=round, red] (0,0)
      --(5,0);
  \end{tikzpicture}
\end{center}
```

This produces the following output:



We can also do a color combination (with the '!') with any color. We can also create an arrow as follows:

Listing 9: An arrow

```
\begin{center}
  \begin{tikzpicture}
    \draw[line width=5mm, cap=round, red, ->] (0,0)
      --(5,0);
  \end{tikzpicture}
\end{center}
```

The `->` means that we are drawing an arrow which looks like this:



There are a variety of arrows that we can draw with `tikz`.^[1]

Here is the code to display another arrow:

Listing 10: Stealth

```
\begin{center}
  \begin{tikzpicture}
    \draw[line width=5mm, cap=round, red!50!brown, >-
      Stealth] (0,0)--(5,0);
  \end{tikzpicture}
\end{center}
```

This produces the following output:



Don't forget to include the `arrows.meta` library while drawing different variants of arrows!

References

- [1] pgf manual. <https://pgf-tikz.github.io/pgf/pgfmanual.pdf#nameddest=subsection.16.3>.